

You are responsible for implementing an auto-throttle controller

- a) Develop an appropriate reduced-state model to which the controller can be added if the linearized equations of motion in wings level, SLF are:

No $\dot{W}$ - derivative

* Maybe tail-less

$$\begin{bmatrix} \dot{u}' \\ \dot{w}' \\ \dot{q}' \\ \dot{\theta}' \end{bmatrix} = \begin{bmatrix} \bar{X}_u & \bar{X}_w & \bar{X}_q & -g \\ \bar{Z}_u & \bar{Z}_w & \bar{Z}_q & 0 \\ \bar{M}_u & \bar{M}_w & \bar{M}_q + U_e & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u' \\ w' \\ q' \\ \theta' \end{bmatrix} \quad [30\%]$$

Steady-
Level -
Flight

- b) What is the simplest controller that would be necessary? Derive the closed-loop system's governing eqns. [40%]
- c) Derive the transfer function for a change in the velocity set-point U^* [30%]

(a) For throttle controller, more likely to activate phugoid mode.

$$\dot{U} = X_u U + X_w W + X_q q - g\theta$$

$$\dot{W} = Z_u U + Z_w W + (Z_q + U_e) q$$

$$\text{where } \dot{\theta} = q, \quad W \rightarrow 0, \quad \dot{W} \rightarrow 0 \quad \text{and} \quad \dot{q} \rightarrow 0$$

$$\therefore \dot{U} = X_u U + X_q \dot{\theta} - g\theta$$

$$0 = Z_u U + (Z_q + U_e) \dot{\theta}$$

$$\therefore \begin{bmatrix} 1 & -X_q \\ 0 & -z_q - Ue \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & -g \\ z_u & 0 \end{bmatrix} \begin{bmatrix} U \\ \theta \end{bmatrix}$$

For throttle controller C_T ,

$$\begin{bmatrix} 1 & -X_q \\ 0 & -z_q - Ue \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} X_u & -g \\ z_u & 0 \end{bmatrix} \begin{bmatrix} U \\ \theta \end{bmatrix} + \begin{bmatrix} X_{C_T} \\ z_{C_T} \end{bmatrix} [C_T]$$

where

$$X_{C_T} = \frac{1}{2} \rho V_\infty^2 S_{ref} \left[\cos(\epsilon + \alpha_e) - \left(\frac{dC_D}{dC_T} \right) e \right]$$

$$z_{C_T} = \frac{1}{2} \rho V_\infty^2 S_{ref} \left[-\sin(\epsilon + \alpha_e) - \left(\frac{dC_L}{dC_T} \right) e \right]$$

(b) PI controller where $u = K_p e(t) + K_I \int_0^t e(\tau) d\tau$

$$e(t) = r - y = r - Cx = r - [1 \ 0]x = U^* - U$$

$$\text{let } \int_0^t e(\tau) d\tau = z \quad \therefore \dot{z} = e(t) = U^* - U$$

$$\therefore u = K_p (U^* - U) + K_I z$$

$$\therefore \begin{bmatrix} 1 & -X_q \\ 0 & -Z_q - U_e \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{\Theta} \end{bmatrix}$$

$$= \begin{bmatrix} X_u & -g \\ Z_u & 0 \end{bmatrix} \begin{bmatrix} U \\ \Theta \end{bmatrix} + \begin{bmatrix} X_{cr} K_p \\ Z_{cr} K_p \end{bmatrix} U^*$$

$$- \begin{bmatrix} X_{cr} K_p \\ Z_{cr} K_p \end{bmatrix} [1 \ 0] \begin{bmatrix} U \\ \Theta \end{bmatrix} + \begin{bmatrix} X_{cr} K_I \\ Z_{cr} K_I \end{bmatrix} Z$$

$$= \begin{bmatrix} X_{cr} K_p & 0 \\ Z_{cr} K_p & 0 \end{bmatrix} \begin{bmatrix} U \\ \Theta \end{bmatrix}$$

$$= \begin{bmatrix} X_u - X_{cr} K_p & -g \\ Z_u - Z_{cr} K_p & 0 \end{bmatrix} \begin{bmatrix} U \\ \Theta \end{bmatrix} + \begin{bmatrix} X_{cr} K_p \\ Z_{cr} K_p \end{bmatrix} U^*$$

$$+ \begin{bmatrix} X_{cr} K_I \\ Z_{cr} K_I \end{bmatrix} Z$$

$$\therefore \begin{bmatrix} 1 & -X_q & 0 \\ 0 & -Z_q - U_e & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \dot{U} \\ \dot{\Theta} \\ \dot{Z} \end{bmatrix}$$

$$= \begin{bmatrix} X_u - X_{cr}K_p & -g & X_{cr}K_2 \\ Z_u - Z_{cr}K_p & 0 & Z_{cr}K_2 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u \\ \theta \\ z \end{bmatrix}$$

$$+ \begin{bmatrix} X_{cr}K_p \\ Z_{cr}K_p \\ 1 \end{bmatrix} U^*$$

$$(c) \quad sMX(s) = AX(s) + BU(s)$$

$$\therefore X(s) = (sM - A)^{-1} BU(s)$$

Derive L_p & N_p derivatives.

$$L = - \int y \, dz'$$

$$N = - \int y \, dx'$$

$X'(y)$ is the sectional force in the x direction
 $Z'(y)$ ——— in the z direction